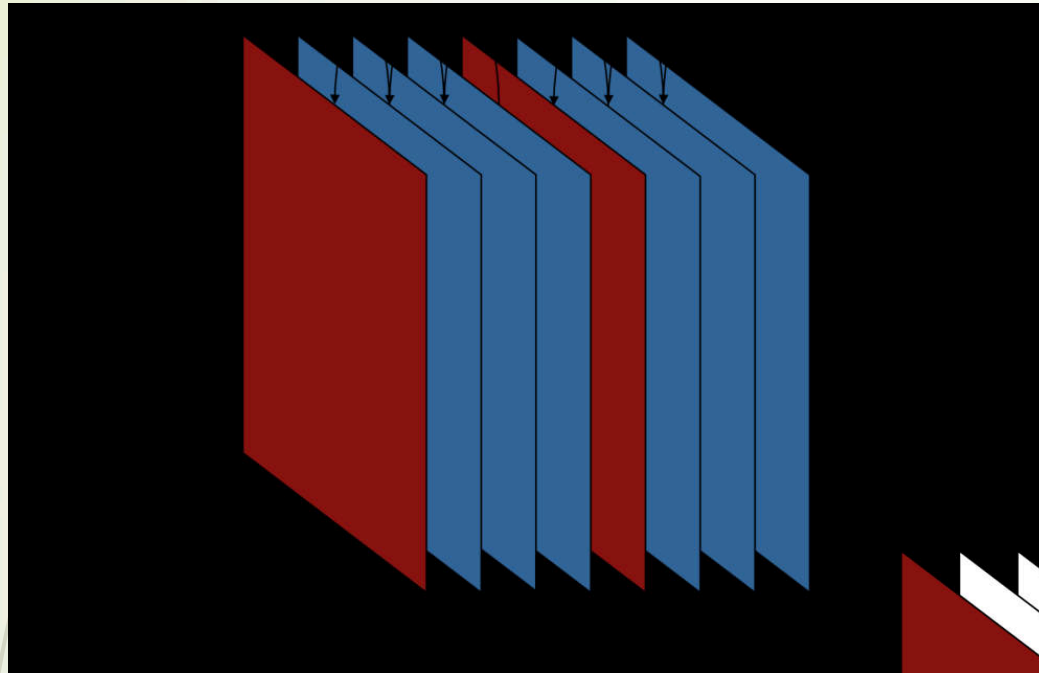
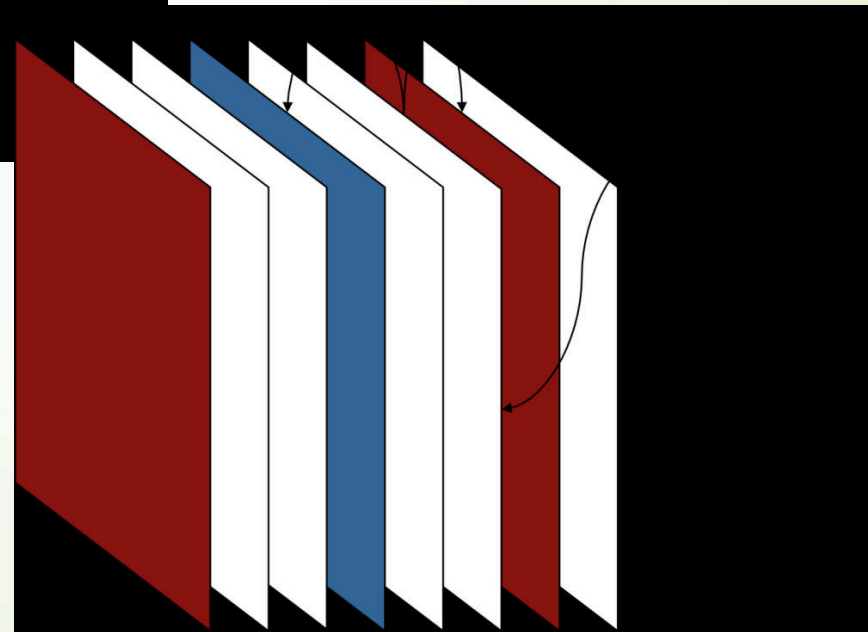


P, B Frames, and Group of Pictures



H.261 (1988)

MPEG-1 (1992)



GoP and Dependency

➤ Example **display order**:

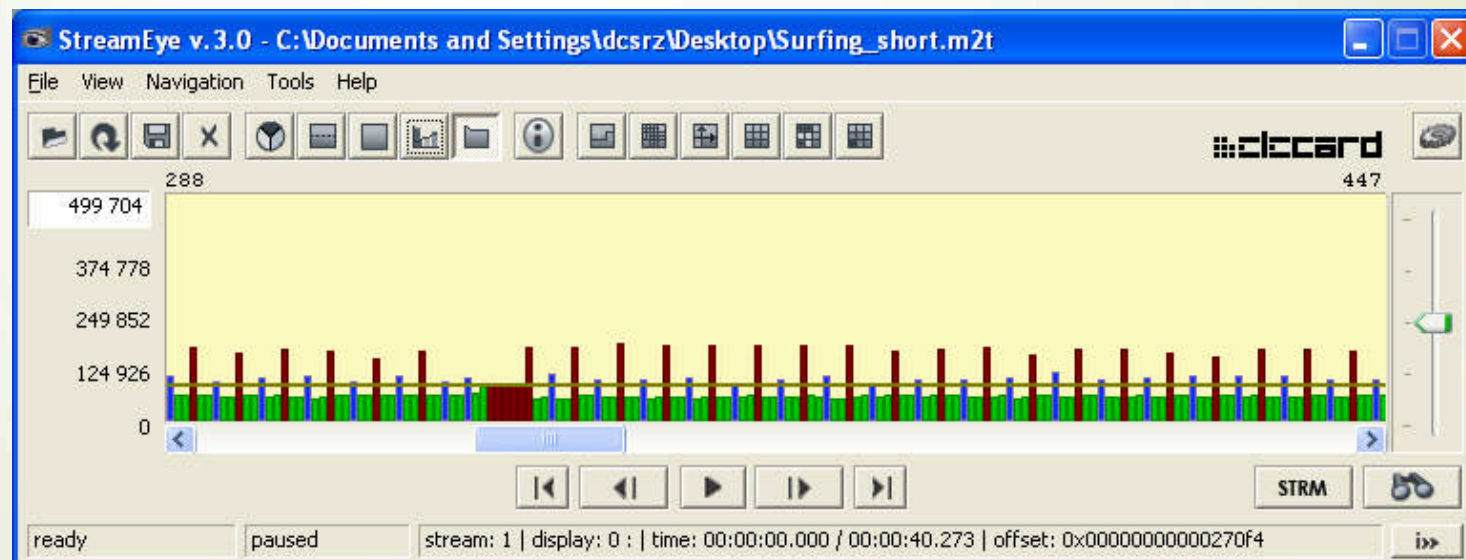
➤ IBBPBBP ...

➤ Example **encoding order**:

➤ IPBBPBB

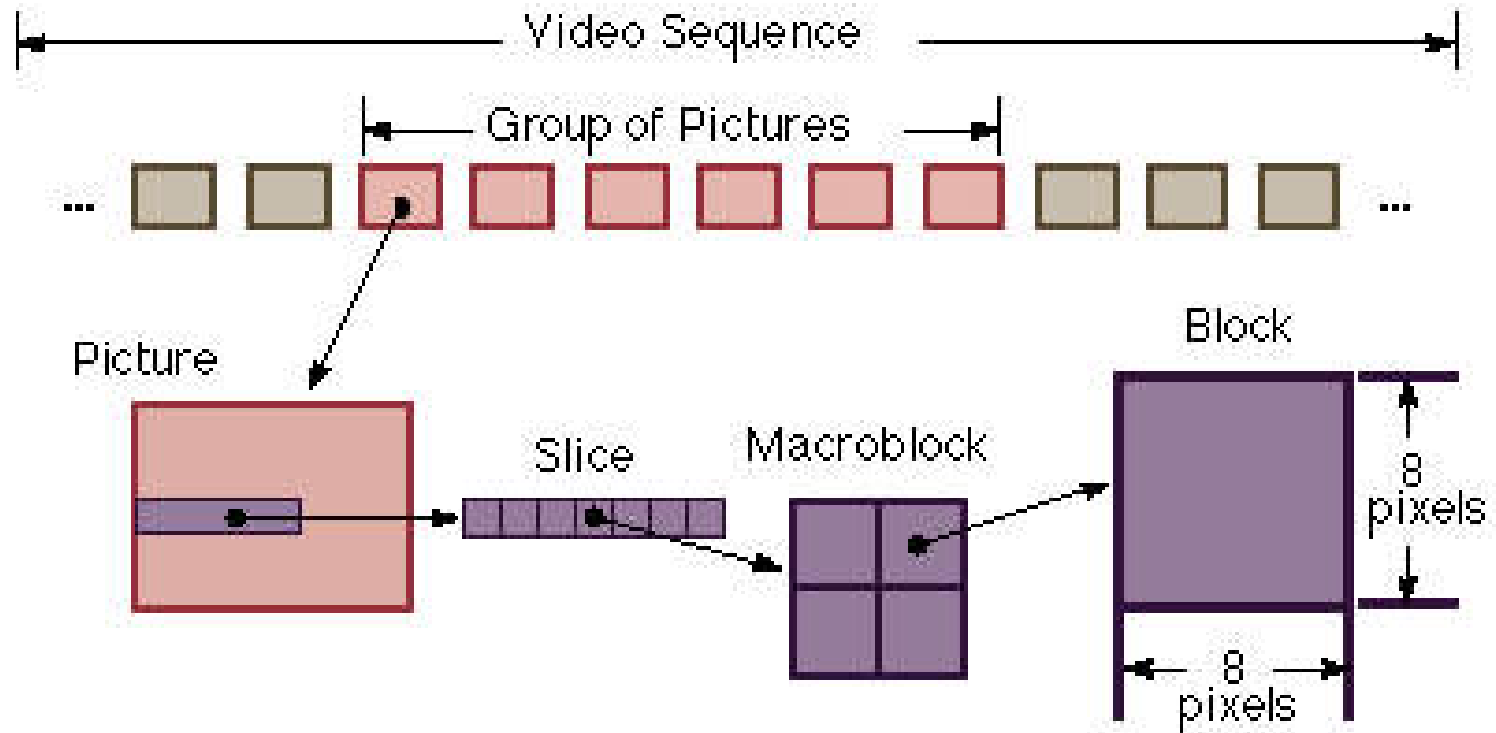
➤ Careful about the difference between them

Frame Type	Typical Ratio
I	10:1
P	20:1
B	50:1



Video Structure

Q: Having more slices in the bitstream allows better **error concealment, but also lead to **lower coding efficiency**. Why?**



I, P, B Settings are at Macroblock Level: B-Frame as an Example

➤ Every macroblock is either

➤ I-macroblock

➤ P-macroblock

Determined by the encoder

➤ B-macroblock

➤ a motion vector + residue wrt a future I/P-frame

➤ 2 motion vectors + residue wrt a previous/future I/P-frame

Video Codecs Are...

discrete cosine transform

- DCT + Quantization + Entropy Coding + **Motion Compensation**
- There are more advanced tools in recent codecs
 - Half-pel motion prediction
 - Skipped macroblock
 - Different sizes of macroblocks
 - Motion vectors across multiple frames
- Key observation: **More compression tools are thrown into more recent codec standards to trade computation for coding efficiency**

Agenda

- Why Compression
- Image Coding Tools
- Video Coding Tools
- Popular Video Codecs

Common Codecs from MPEG

MPEG Standards	Bit-rate	Usage
MPEG-1	1.5Mbps	VCD
MPEG-2	3-45 Mbps	DVD, SVCD, HDTV
MPEG-4/ H.264/AVC	Scalable, 1/2 MPEG-2	QuickTime, DivX, AVCHD, Cable TV, YouTube, ...
H.265/HEVC	Scalable, 1/2 H.264	New generation, 4K content
"H.266"	Scalable, 1/2 H.265	Next generation, 8K content

Other Codecs: VP9 and AV1

- VP9 is an open and royalty free video coding format developed by Google
 - Same concepts/tools compared to MPEG codecs
- Successor to VP8
- Supported by most web browsers (except Safari)
- Used by YouTube
- Successor: AV1

Try A Few Public Reference Codecs

- Download source code, compile and play with
 - ffmpeg
 - mpeg_stat
 - jsvm
 - shm
- Find raw video sequences in YUV420 format
- Try different encoding parameters

How Codecs Affect Media Streaming?

- How to package data into packets? ← **packtization**
- How to deal with packet loss? ← **error detection, error recovery, error concealment**
- How to deal with bursty traffic? ← **rate control**
- How to predict decoding time? ← **real-timeness**
- :

Take-Away Messages

- Compression removes data for which the human visual system is not sensitive.
- Current codecs are based on DCT, quantization, entropy coding, and motion compensation
 - Motion compensation, motion estimation (or prediction), motion vector,
- Codec standards are essentially defined by their bitstreams (not encoders)
- Container formats (MPEG-2 TS, ISO/BMFF) are important for system designers ← didn't cover

Questions

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